

YIFAN ZHANG

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EDUCATION

Johns Hopkins University

Ph.D. in Biostatistics

Zhejiang University

B.S. in Statistics

Maryland, USA

2026-present

Zhejiang, China

2022-2026

RESEARCH EXPERIENCE

School of Medicine and Public Health, University of Wisconsin-Madison

Research Assistant

Wisconsin, USA

Advisor: Prof. Guanhua Chen

Compositionality-Aware Cross-Study Microbiome Batch Correction to Improve Phenotype Prediction

09/2025-Present

- Developing a compositionality-aware cross-study microbiome debiasing framework.
- Designed for application to multi-cohort Alzheimer's disease microbiome datasets.
- Preliminary experiments demonstrate improved robustness and stability over existing microbiome batch correction approaches across metagenomic and 16S rRNA sequencing datasets.

School of Public Health, Yale University

Summer Research Intern

Connecticut, USA

Advisor: Prof. Hongyu Zhao

Iterative Approach for Spatial Transcriptomics Alignment and Deconvolution

03/2025-Present

- Designed and implemented an iterative framework for Spatial Transcriptomics (ST) alignment and deconvolution integrating graph attention networks and Optimal Transport (OT).
- Deconvoluted cell-type proportions serve as biologically meaningful guidance for the OT process.
- Consistently improved alignment accuracy over state-of-the-art methods on the Dorsolateral Prefrontal Cortex (DLPFC) benchmark dataset.
- Extending the framework to large scale high-resolution ST datasets and assessing its stability under near single-cell spatial resolution.

School of Mathematical Sciences, ZJU

Research Assistant

Zhejiang, China

Advisor: Prof. Kewei Liang

Multimodal Deep Learning Model for Laryngeal and Hypopharyngeal Lesions Risk Prediction

03/2024-11/2025

- Contributed to building a multimodal deep-learning pipeline that integrates laryngoscopic images, patient chief complaints and demographic features for continuous risk scoring of laryngeal and hypopharyngeal lesions.
- Designed and conducted clinician-machine comparison experiments, demonstrating diagnostic performance surpassing senior laryngologists on three independent validation centers.

PUBLICATIONS

Preprints

Zhang, Jiahong and Liang, Kewei and Yao, Qingxiu and **Zhang, Yifan** and Mo, Qun and Chen, Xiaojing and Qu, Siyuan and Luo, Yunzhen and Chen, Hengchao and Dai, Libo and Zhou, Shui-Hong, A Multimodal Deep Learning Model for Laryngeal and Hypopharyngeal Lesions Diagnosis: A Multicenter Retrospective Study. Available at SSRN: <https://ssrn.com/abstract=5216153> or <http://dx.doi.org/10.2139/ssrn.5216153>

TEACHING EXPERIENCE

Teaching Assistant

MATH3404M Statistical Computing, Zhejiang University

Spring-Summer, 2026

SELECTED HONORS & AWARDS

- Lingjun Leading Scholarship, 2025
- 2nd Prize of Zhejiang University Scholarship, 2023 & 2024
- 3rd Prize of Higher Mathematics Competition for College Students in Zhejiang Province, 2023
- 1st Prize of the 15th National College Students Mathematics Competition (A Class), 2023
- 2nd Prize in Physics Innovation (Theory) Contest for College Students of Zhejiang Province, 2023
- 3rd Prize in 2023 National English Competition for College Students, 2023
- 1st Prize of the 11th English Writing Competition for College Students, Zhejiang University, 2023
- 2nd Prize of the 19th English Speech Contest for College Students, Zhejiang University, 2023
- 2nd Prize in the 8th English Reading Contest for College Students of Zhejiang University, 2023